
Association HTTP method ↔ CRUD

- **GET** → read
- **POST** → create
- **PUT** → update
- **DELETE** → delete
Additional method
- **PATCH** → partial update
- **OPTIONS** → available http methods

MongoDB CRUD API

- **USERS**
 - **Create**
 - Create new use
POST /api/users/register
 - **POST /api/users/login**
 - **Read**
 - get user by id
GET /api/customers/me
GET /api/admin/customers/{id}
 - get user lists
GET /api/admin/customers
 - get user wallet (cash and blocked cash)
GET /api/customers/me/cash
 - get user portfolio (the wallets with owned assets)
GET /api/customers/me/portfolio
 - get user lastTransaction (last 10 transactions)
GET /api/customers/me/lastTransactions
 - **Update**
 - update user by id (only account information)
PUT /api/customers/me/account
 - suspend user (for violation, used by admins)
PUT /api/admin/customers/{id}/suspend
 - unsuspend user
PUT /api/admin/customers/{id}/unSuspend
 - **Delete**
 - delete user (soft delete)
DELETE /api/admin/users/{id}
- **TRANSACTIONS**
 - **Create**
 - Create new transaction (purchase/sell/deposit/withdrawal)
POST /api/customers/transactions
 - **Read**
 - get transaction by transactionId
GET /api/admin/transactions/{id}

- get all transactions of a user
GET /api/customers/me/transactions
 - GET /api/admin/users/{userId}/transactions
 - get transactions by:
GET /api/customers/transactions?status=&type=&from=&to=
GET /api/admin/transactions?status=&type=&from=&to=&userId=
- Update
 - update transaction status
PUT /api/admin/transactions/{id}
- Delete
 - delete transaction (soft delete)
DELETE /api/admin/transaction/{id}
- ASSETS
 - Create
 - Insert new asset
POST /api/admin/assets
 - Read
 - get asset by Symbol
GET /api/users/assets/{symbol}
GET /api/customers/assets/{symbol}
GET /api/admin/assets/{symbol}
 - get assets by type
GET /api/users/assets/type/{type}
GET /api/customers/assets/type/{type}
GET /api/admin/assets/type/{type}
 - get all assets
GET /api/users/assets
GET /api/customers/assets
GET /api/admin/assets
 - Update
 - update asset metadata
PUT /api/admin/assets/{symbol}
 - Delete
 - delete asset (soft delete)
DELETE /api/admin/assets/{symbol}
- ASSET_PRICES
 - Create
 - insert price_assets
POST /api/admin/asset-prices
 - Read
 - get prices by symbol and date
GET /api/customers/asset-prices/{symbol}?from=start&to=end
GET /api/admin/asset-prices/{symbol}?from=start&to=end
 - get latest price
GET /api/customers/asset-prices/{symbol}/latest
GET /api/admin/asset-prices/{symbol}/latest
 - Update
 - Update an asset_prices
PUT /api/admin/asset-prices/{symbol} -> NO IMPLEMENTATO

- **Delete**
 - delete price
DELETE /api/admin/asset-prices/{symbol}
- **NEWS**
 - **Create**
 - insert new news
POST /api/admin/news
 - **Read**
 - Get list of news (all news):
GET /api/users/news
GET /api/customers/news
GET /api/admin/news
 - Get list of news (from offset to limit, if are first 10 use Redis cache)
GET /api/users/news/latest?offset=0&limit=10
GET /api/customers/news/latest?offset=0&limit=10
GET /api/admin/news/latest?offset=0&limit=10
 - Get list of news by sector
GET /api/customers/news/sector/{sector}
GET /api/admin/news/sector/{sector}
 - Get list of news by sector (from offset to limit, if are first 10 use Redis cache)
GET /api/customers/news/latest/sector/{sector}?offset=0&limit=10
GET /api/admin/news/latest/sector/{sector}?offset=0&limit=10
 - get news by id
GET /api/customers/news/{id}
GET /api/admin/news/{id}
 - **Update**
 - Update a news
PUT /api/admin/news/{id}
 - **Delete**
 - delete news by id
DELETE /api/admin/news/{id}
- **Mixed Redis API (current prices)**
 - Get current price for an assets: get the current price of an asset if is available in Redis
GET /api/users/assets/asset_current_price/{symbol}
GET /api/customers/assets/asset_current_price/{symbol}
GET /api/admin/assets/asset_current_price/{symbol}

MongoDB AGGREGATION API

- **User**
 - View the 10 users with the largest portfolios in terms of different assets.
GET /api/admin/analytics/users/top-variety
 - View the 10 users with the largest amount of a given asset in their portfolio.
GET /api/admin/analytics/users/top-holders
 - View the average, minimum and maximum number of distinct assets held by users.
GET /api/admin/analytics/users/global-stats
 - Total number of registered or suspended or unsuspend users (pure analytics, not in the aggregation)
GET /api/admin/analytics/users/user-status-stats

- Total number of registered or suspended or unsuspend users (pure analytics, not in the aggregation)
GET /api/admin/analytics/users/new-users
- **Asset**
 - Number of assets by type (share / ETF / crypto)
GET /api/customer/analytics/assets/asset-type-count
GET /api/admin/analytics/assets/asset-type-count
 - Top 10 sectors by number of listed share
GET /api/customer/analytics/assets/top-asset-sector
GET /api/admin/analytics/assets/top-asset-sector
- **Asset_prices**
 - View the top 10 assets with the best growth last day/week/month/year.
GET /api/customer/analytics/assets-prices/top-growt-asset?window=
GET /api/admin/analytics/assets-prices/top-growt-asset?window=
 - View the top 10 assets with the worst decline last day/week/month/year.
GET /api/customer/analytics/assets-prices/worst-decline-asset?window=
GET /api/admin/analytics/assets-prices/worst-decline-asset?window=
 - See the 10 assets that have consistently raisen over the past week (i.e., each day closed higher than its opening for a week) and their average daily growth rate.
GET /api/customer/analytics/assets-prices/top-stable-raisen
GET /api/admin/analytics/assets-prices/top-stable-raisen
 - See the 10 assets that consistently fell last week (i.e., closed every day negatively compared to their opening for a week) and their average decline loss ratio.
GET /api/customer/analytics/assets-prices/worst-stable-fell
GET /api/admin/analytics/assets-prices/worst-stable-fell
- **Transaction**
 - View the top 10 most traded assets on the platform for day/week/month/year.
GET /api/customer/analytics/transactions/most-traded
GET /api/admin/analytics/transactions/most-traded
 - Analyze the number of transaction, by categories or by payment type for day/week/month/year. Offering an overview of user payment preferences [interessante per controllo dell'admin]
GET /api/admin/analytics/transactions/number-transaction?type=
 - View the total amount of money invested in assets over a specific time range (day, week, month, ...). Sum of totalPrice of all buy transactions within a given time window. [interessante per controllo dell'admin]
GET /api/admin/analytics/transactions/invested-money/{symbol}?window=
 - View the Top-K users with the highest net financial flow generated by buy and sell operations within a selected time window (day. week, month, ...).
Buy operations contribute negatively to the flow, while sell operations contribute positively. Users are ranked in descending order based on their net financial impact.
GET /api/admin/analytics/transactions/top-net-flow?window=
 - View the Top-K users with the worst net financial flow generated by buy and sell operations within a selected time window (day, week, month, ...).
This aggregation highlights users who have invested the largest amount of capital in asset purchases relative to their sell operations.
GET /api/admin/analytics/transactions/worst-net-flow?window=
- **News**

- See the number of news per Sector in a time window.
GET /api/admin/analytics/news/trending-sectors?window=
- Identify the top 5 companies that have been mentioned in the highest number of news articles within a given time window.
GET /api/admin/analytics/transactions/top-mentioned-companies?window=

Key-Value: CACHE Operations

Operations to be modified in the APIs to integrate Redis. These operations affect the data stored within the Redis cache layer.

Assets

- **When an asset is created:**
 - Add the asset to the **asset list by type**.
- **When an asset is deleted:**
 - Remove it from the **asset list by type**.
 - Delete its **current price** key.
 - Process and remove the **intraday prices** for that asset.
 - Remove it from the **most/best/worst traded** lists.
- **When an asset is reactivated:**
 - Add the asset back to the **asset list by type**.

Users

- **Register:** Load user data into Redis.
- **Login:** Load user data into Redis.
- **Logout:** Delete (evict) user data from Redis.
- **Read (User Key):** Update the **TTL** (Time To Live) for all keys related to that user.
- **Update (User Key):** Update the **TTL** for all keys related to that user.
- **When transactions are created or executed:** Update or reset the **user cache**.
- **When a user is deleted/suspended:** Clear the cache for that specific user.

News

- **When a new article is added:**
 - Update news:latest (Global list).
 - Update news:latest:sector:{sector} (Sector-specific list).
 - Add/Update the individual news:{id} hash.
- **When a news article is deleted:**
 - Check if the ID exists in any of the lists (Global or Sector) and remove it if present.